
MAGE: Learning an Effective Theory of Superconductivity from Composition Alone

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Abstract

Superconductor discovery is limited by experimental cost, data incompleteness, and unresolved theoretical disagreements across material families. The long-sought goal of ambient-pressure high-temperature superconductivity remains elusive, and the incompleteness of current theory leaves open whether it is physically achievable. I present MAGE (Materials Autoencoder for Generative Exploration), a multi-task encoder-decoder that maps chemical formulas into a 2048-dimensional latent space while jointly optimizing formula reconstruction, T_c prediction, SC/non-SC classification, family classification, Magpie feature reconstruction, and stoichiometry prediction. Trained on 56,313 materials (26,191 superconductors; 30,122 non-superconductors), MAGE achieves $R^2 = 0.9999$ on T_c prediction (MAE = 0.12 K)—a $27\times$ error reduction over a Magpie-only baseline ($R^2 = 0.920$). On 45 held-out superconductors never seen during training, a Z-space search generating $\sim 109,000$ candidates per target recovers exact compositional matches for 17/45 targets (37.8%) and the correct element set for 43/45 (95.6%), with zero hallucinated elements, 100% correct SC classification, median T_c error of 0.40 K, and 98% of residual errors limited to stoichiometric fraction imprecision—all without receiving any target information. By comparison, roundtrip encode-decode achieves only 3/45 exact matches (mean similarity 0.944). Analysis of the learned latent space reveals family-specific gradient profiles, physically meaningful clustering, and a novel geometric diagnostic—*latent torsion*—that identifies which materials require structural information beyond composition for accurate T_c prediction. The model has learned what I characterize as a 2048-parameter effective theory of superconductivity spanning 14 material families, implicitly encoding the diverse physical mechanisms that give rise to superconductivity within a single unified representation.

1 Introduction

Discovering new superconducting materials remains a central challenge in condensed matter physics. Since Onnes’ discovery of superconductivity in mercury in 1911, the field has progressed through elemental superconductors, intermetallic compounds (Nb_3Sn , Nb_3Ge), the cuprate revolution beginning with Bednorz and Müller’s 1986 discovery of La-Ba-Cu-O (44), iron pnictides (41), MgB_2 (40), and most recently high-pressure hydrides where H_3S achieves $T_c = 203$ K at 155 GPa (45) and LaH_{10} reaches 250–260 K at 170–200 GPa (46). Each major family was discovered largely through serendipity or chemical intuition rather than systematic prediction—a pattern noted across the history of the field (48).

Superconductivity presents two intertwined holy grails. The *applied* goal is high-temperature superconductivity at ambient pressure, which would transform energy transmission, computing, and transportation. The *theoretical* goal is finding the equations that unify the diverse physical mechanisms giving rise to superconductivity. A unifying description *must* exist: all superconductors, regardless of mechanism, exhibit the same empirical phenomena—zero electrical resistance, the

Meissner effect, and flux quantization in units of $h/2e$. These universal properties demand a universal theoretical framework. The profound challenge is that the microscopic routes to these shared phenomena are remarkably diverse—phonon-mediated BCS pairing (35), d -wave spin-fluctuation pairing in cuprates (36), $s_{\pm}$ -wave inter-band pairing in iron pnictides (41), two-gap strong coupling in MgB_2 (40), and heavy-fermion/unconventional pairing—spanning vastly different energy scales, symmetries, and chemistry. These two goals are deeply connected: without a complete theory, it is unknown whether ambient-pressure room-temperature superconductivity is physically achievable or whether fundamental constraints—analogue to the $T_c/J \approx 0.04\text{--}0.07$ bound observed across unconventional superconductors (49)—impose an upper limit that current materials are already approaching. A complete theory unifying these mechanisms would have implications far beyond materials science. Superconductivity has already had one such transformative impact: Anderson’s demonstration that the photon acquires an effective mass inside a superconductor through spontaneous symmetry breaking (47) served as the direct prototype for the Higgs mechanism, which explains how gauge bosons acquire mass in the Standard Model of particle physics. A deeper understanding of what connects the diverse manifestations of superconductivity—the universal principle that the empirical phenomena demand—could yield similarly far-reaching theoretical insights.

A single model trained across all known superconductor families is, implicitly, an attempt to learn this unifying physics from data. If the model organizes its latent space into a manifold where diverse mechanisms coexist with smooth transitions and shared geometric structure, it provides evidence—not proof, but evidence—that the superficially different microscopic mechanisms are connected by deeper commonalities that a sufficiently expressive representation can capture. The combinatorial space of possible compositions exceeds 10^{10} candidates, motivating computational approaches that can learn the implicit physics governing superconductivity and use it to navigate this space efficiently.

Materials informatics has made significant progress in predicting critical temperature (T_c) from chemical composition using machine learning. Hamidieh (1) demonstrated XGBoost models on SuperCon data; Stanev et al. (3) achieved $R^2 \approx 0.88$ with random forests and Magpie features (2); more recent work by Jung et al. (9) reached $R^2 = 0.952$ using gradient-boosted feature selection. However, these composition-only methods face a fundamental ceiling: a chemical formula is an incomplete specification of a superconductor. Many variables that critically determine T_c —crystal structure, oxygen ordering, apical oxygen distances, pressure, defect chemistry—are absent from composition-only databases (7). In cuprates, T_c can vary by up to a factor of 10 between materials with the same in-plane electronic structure but different out-of-plane geometry (15; 17). The SuperCon database, the largest repository of superconductors, records only chemical formulas; structural data is sparsely recorded and full 3D crystal structures are never given (7).

Generative approaches have begun to address the inverse design problem—proposing new candidate materials rather than merely predicting properties of known ones. Wines et al. (11) combined a crystal diffusion variational autoencoder (CDVAE) with ALIGNN to generate and screen superconductor candidates. Zhong et al. (12) used a VAE-GAN hybrid for high- T_c composition generation. ScGAN (13) achieved a 23-fold improvement in superconductor discovery rate over random search. However, these approaches are typically two-stage pipelines: one model generates candidates, a separate model screens them for properties. The generative and predictive capabilities are architecturally disconnected.

In this work, I present MAGE, a unified multi-task encoder-decoder (4) where the same 2048-dimensional latent space simultaneously supports formula generation and property prediction through shared decoder pathways. The architecture introduces several innovations: (1) a three-branch encoder combining learned element representations, Magpie compositional features, and T_c conditioning; (2) a pointer-generator decoder with fraction-aware tokenization for precise stoichiometric generation; (3) six jointly optimized decoder heads that create self-consistency learning channels shaping the latent space; and (4) a feedback mechanism where predicted T_c conditions the formula decoder through cross-attention memory tokens. I train on 56,313 materials with a deliberately balanced SC/non-SC split for class-balanced training, achieving $R^2 = 0.9999$ on T_c prediction and demonstrating generalization on 45 held-out superconductors across 9 families through both roundtrip reconstruction and systematic latent space search. I analyze the learned latent representation to show that the model has organized its 2048-dimensional space into a physically meaningful effective theory of superconductivity.

2 Related Work

Composition-based T_c prediction. Machine learning approaches to T_c prediction have progressed from random forests on Magpie features ($R^2 \approx 0.88$ (3)) through XGBoost ($R^2 \approx 0.92$ (1)) to gradient-boosted feature selection ($R^2 = 0.952$ (9)). Including structural information via SOAP descriptors improves performance to $R^2 = 0.929$ (10), but requires crystal structure data that is unavailable for most superconductors. These methods predict properties of existing materials but do not generate new ones.

Generative models for superconductors. Several generative approaches have been proposed: CDVAE with ALIGNN screening (11), VAE-GAN hybrids (12), ScGAN (13), SuperDiff (14), and GANs for crystal structure prediction (16). These are two-stage pipelines where generation and property prediction are handled by separate models. This work differs by unifying generation and prediction in a single architecture where the latent space serves both functions simultaneously.

Sequence-based chemical generation. The Chemical VAE (6) demonstrated that joint training of a VAE with a property predictor produces latent spaces organized by property values. Materials-specific generative models have since been developed for solid-state inverse design (29) and composition space sampling (30). The pointer-generator mechanism (5) enables copying tokens from input sequences, which I adapt for exact element symbol reproduction. The fraction-aware tokenization extends these approaches to handle the decimal and fractional stoichiometries common in inorganic superconductors.

3 Dataset

3.1 Training Data: 56,313 Materials

The model is trained on a composite dataset substantially larger and more diverse than the SuperCon subset used in prior work. Table 1 summarizes the data sources.

Table 1: Training dataset composition

Component	Source	Count
SuperCon SC	NIMS (8)	16,521
NEMAD SC	NEMAD (27)	6,930
JARVIS Hydrides SC	JARVIS-DFT (34)	2,740
Total SC		26,191
Materials Project non-SC	MP (28)	14,040
NEMAD non-SC	Magnetic/thermo/aniso	9,154
MP Targeted non-SC	Rare element neighborhoods	3,500
Total non-SC		30,122
Grand Total		56,313

All materials pass through a uniform pipeline: pymatgen (19) parsing, 145 Magpie features via matminer (26), and 6 L_p -norm features, producing consistent feature vectors per sample.

3.2 Data Pipeline

The pipeline consists of three stages. Stage 1 parses 16,521 superconductor entries from NIMS SuperCon, converting decimal formulas to fraction notation (e.g., $\text{YBa}_2\text{Cu}_3\text{O}_{6.9}$ to $\text{Y}(1/1)\text{Ba}(2/1)\text{Cu}(3/1)\text{O}(69/10)$), computing 145 Magpie features, and assigning superconductor family categories. Stage 2 ingests 19,058 raw NEMAD entries, filters to 15,339 experimental-only records, extracts T_c , normalizes formulas through pymatgen, deduplicates against SuperCon by canonical composition, and excludes the 45-sample holdout set. Stage 3 loads non-superconductors from 4 sources (NEMAD magnetic, thermoelectric, anisotropy, and Materials Project), deduplicates across sources, and removes any compositions matching known superconductors or the holdout set.

All deduplication uses canonical pymatgen compositions, making the pipeline robust to formatting differences. Feature schemas are identical across all outputs (151 base columns).

3.3 Why Include Non-Superconductors?

The non-SC materials serve three purposes developed iteratively during training:

Chemical vocabulary expansion. Superconductor formulas cluster around certain elements (Cu, O, Ba, Sr, La, Fe, As) and stoichiometric patterns. Early in training, the formula decoder struggled to reconstruct compositions with rare elements or unusual doping fractions. Non-SC materials from Materials Project and NEMAD expose the encoder and decoder to broader chemical vocabulary. A targeted batch of 3,500 MP entries from rare element neighborhoods (Tl+Li, Hg+Ce, Gd+Pb) was added later in training specifically to address reconstruction failures on holdout-relevant compositions. The NEMAD integration was particularly impactful for underrepresented elements: NEMAD provided 95.1% of all hydrogen-containing SC samples (405 of 426 total) and introduced argon as an entirely new element not present in the original SuperCon dataset.

Class balance. The approximately 46/54 SC/non-SC split (26,191 vs. 30,122) prevents the SC classifier from learning a trivial majority-class prior. Training on all SC with few non-SC teaches “everything is superconducting”; training on all known compounds teaches “nothing is superconducting.”

Phase boundary geometry. Non-SC materials force the latent space to encode the SC/non-SC phase boundary as a geometric feature, providing gradient signal for the SC classifier head.

3.4 Holdout Set

45 superconductors (5 per family $\times$ 9 families) are drawn from the test split (seed=42) and completely excluded from training. Table 2 summarizes the holdout composition.

Table 2: Holdout set composition by family

Family	T_c Range (K)	Representative
YBCO	93.8–106.5	$\text{Y}_3\text{Ba}_5\text{Cu}_8\text{O}_{18}$
LSCO	81.0–121.6	$\text{La}_1\text{Ba}_2\text{Cu}_3\text{O}_{137/20}$
Hg-cuprate	128.7–143.0	$\text{Hg}_{33/50}\text{Pb}_{17/50}\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}$
Tl-cuprate	110.0–123.5	$\text{Tl}_2\text{Ba}_2\text{Ca}_2\text{Cd}_{1/10}\text{Cu}_3\text{O}$
Bi-cuprate	110.0–136.0	$\text{Bi}_{9/5}\text{Pb}_{1/5}\text{Sb}_{1/10}\text{Sr}_2\text{Ca}_2\text{Cu}_2\text{O}$
Iron-based	46.8–53.0	$\text{Sm}_{13/20}\text{Th}_{7/20}\text{Fe}_1\text{As}_1\text{O}_1$
MgB ₂	35.3–38.0	$\text{Mg}_{97/100}\text{Na}_{3/100}\text{B}_2$
Conventional	20.4–42.0	$\text{Nb}_3\text{Al}_{71/100}\text{Ge}_{29/100}$
Other	115.0–132.0	$\text{Ba}_2\text{Ca}_5\text{Cu}_7\text{O}_{14}$

4 Methods

4.1 Architecture Overview

The MAGE ($\sim 115\text{M}$ parameters: 7.2M encoder, 107.6M decoder) encodes chemical formulas into a 2048-dimensional latent vector $\mathbf{z}$ and decodes through a shared backbone into six task-specific heads. Fig. 1 shows the architecture. The encoder is dominated by auxiliary heads (SC classifier, competence, fraction predictor, family classifier, Magpie reconstruction) plus the fusion layers. The decoder is dominated by the Transformer decoder (50.4M, 47%) and the latent-to-memory projection (42.0M, 39%) which maps the 2048-dim latent $\mathbf{z}$ into 16 memory tokens for cross-attention.

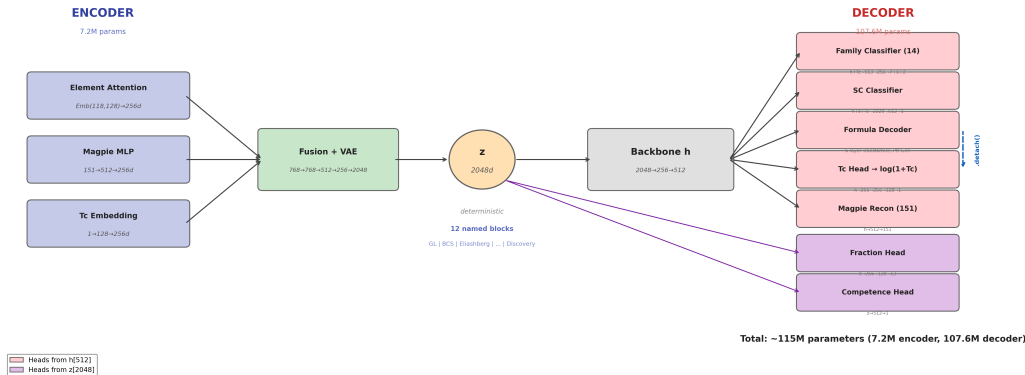


Figure 1: MAGE architecture. Three encoder branches (element composition, 151 compositional features, T_c) fuse into a 2048-dimensional deterministic latent vector $\mathbf{z}$, partitioned into 12 named physics coordinate blocks. The shared backbone $\mathbf{h}$ [512] feeds five decoder heads (pink); two auxiliary heads (fraction, competence; purple) branch directly from $\mathbf{z}$. The T_c head conditions the formula decoder via cross-attention memory tokens with `.detach()` (dashed), creating dual supervision without gradient leakage. Total: $\sim 115\text{M}$ parameters (7.2M encoder, 107.6M decoder).

Encoder. A three-branch Full Materials Encoder processes: (1) the tokenized chemical formula through learned element embeddings (Embedding(118, 128) $\rightarrow$ 8-head self-attention $\rightarrow$ 256-dim), (2) the 151-dimensional compositional feature vector (145 Magpie + 6 L_p -norm) through an MLP (151 $\rightarrow$ 512 $\rightarrow$ 256 with GELU), and (3) the scalar T_c value through a conditioning branch (1 $\rightarrow$ 128 $\rightarrow$ 256 with GELU). The three 256-dim branch outputs are concatenated (768-dim), fused through Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU, and projected to $\mathbf{z} \in \mathbb{R}^{2048}$ via μ and σ heads.

Shared decoder backbone. $\mathbf{z}$ is processed through two shared layers:

$$\begin{aligned} \mathbf{z}[2048] &\rightarrow \text{Linear}(2048, 256) \rightarrow \text{LN} \rightarrow \text{GELU} \\ &\rightarrow \text{Linear}(256, 512) \rightarrow \text{LN} \rightarrow \text{GELU} \rightarrow \mathbf{h}[512] \end{aligned} \quad (1)$$

All six decoder heads branch from the shared hidden state $\mathbf{h}$.

Decoder heads: (1) *Autoregressive formula decoder*: a 6-layer Transformer with 16 memory tokens, using a pointer-generator with copy mechanism and fraction-aware tokenization, reconstructing formulas token-by-token. Hungarian matching (25) is used for parallel slot-based prediction, eliminating autoregressive ordering constraints during training; (2) *T_c prediction head*: projection through a residual block to normalized $\log(1 + T_c)$ space; (3) *SC/non-SC classifier*: binary classification; (4) *Family classifier*: 14-class hierarchical classification; (5) *Magpie reconstruction head*: reconstructs 151 compositional features; (6) *Stoichiometry head*: predicts element counts and fractions.

T_c as decoder conditioning signal. The predicted T_c , SC probability, family classification, and element count are projected into 4 cross-attention memory tokens (24) (24 $\rightarrow$ 4096 $\rightarrow$ 4 $\times$ 1024 dims) that the formula decoder attends to at every generation step. During training, these predictions are passed with `.detach()`, preventing gradient backpropagation through the T_c head from the decoder but ensuring the shared backbone $\mathbf{h}$ must serve both tasks simultaneously. This creates a dual supervision mechanism: the T_c head is trained by its own regression loss, and it must also produce conditioning signals that help the formula decoder—if the T_c prediction is poor, the formula decoder loses a useful attention signal. Importantly, the T_c head is only active for materials classified as superconducting; when the SC classifier predicts non-SC, the T_c head is bypassed and a fill-in value of 0 K is assigned. This means that a reported T_c of 0.0 K in the evaluation results indicates an SC classification failure, not a T_c prediction of zero.

4.2 The 2048-Parameter Effective Theory

The latent space dimensionality is a deliberate design choice driven by the incompleteness of the input data. A typical superconductor formula is specified by 8–12 numbers (element identities and stoichiometries); the Magpie representation expands this to 145 features. The encoder maps every material into 2048 dimensions—roughly $200\times$ larger than the raw compositional specification.

This expansion provides representational capacity for *hidden variable inference*. The encoder cannot observe crystal structure, oxygen ordering, or apical oxygen distances, but if certain compositions consistently appear with certain structural motifs, the encoder can learn statistical proxies from compositional patterns. Importantly, selectively including structural data where available would introduce systematic training bias, since structural information is well-characterized for some families (BCS metals, some cuprates) but sparse for others (organics, heavy-fermion, many doped variants). Using composition uniformly across all families avoids this inhomogeneity.

The 2048 dimensions are partitioned into 12 named coordinate blocks (Table 3). Blocks 1–2 (Ginzburg-Landau, BCS) are constrained by physics self-consistency losses. Block 8 (compositional) is actively supervised against 15 formula-derivable targets. Block 11 (Magpie encoding) learns a linear projection of the 145 Magpie features into 62 latent dimensions. Block 12 (dims 512–2047) is entirely unsupervised discovery space.

Table 3: Physics Z coordinate blocks

Block	Name	Dims	Status
1	Ginzburg-Landau	0–19	Self-consistency
2	BCS/Microscopic	20–49	Self-consistency
3	Eliashberg	50–69	Placeholder
4	Unconventional	70–109	Placeholder
5	Structural	110–159	Placeholder
6	Electronic	160–209	Placeholder
7	Thermodynamic	210–269	Placeholder
8	Compositional	270–339	Active
9	Cobordism	340–399	Self-consistency
10	Dim. Ratios	400–449	Self-consistency
11	Magpie Encoding	450–511	Active
12	Discovery	512–2047	Unsupervised

The six decoder heads create self-consistency learning channels: the family classifier forces $\mathbf{z}$ to encode family-discriminating features; the SC classifier forces the phase boundary; the Magpie head preserves compositional statistics. The T_c head operates on a latent space already shaped by these competing constraints. I call the resulting representation a *2048-parameter effective theory of superconductivity*—compare with BCS/Allen-Dynes (32) (3 parameters: λ , $\omega_{\log}$, μ^* ; applicable to one family) or the Presland dome (33) (3 parameters: $T_{c,\max}$, p_{opt} , dome width; applicable to one cuprate sub-family).

4.3 Training

The model is trained with Adam ($\text{lr} = 10^{-4}$, batch size 1024) on a single NVIDIA A100 80GB GPU via Google Colab, implemented in PyTorch (18). Despite the historical “VAE” nomenclature in the codebase, KL divergence is weighted at 10^{-4} —effectively zero—making the model a *deterministic autoencoder* rather than a variational model. This was a deliberate choice: the generative capability comes from the autoregressive formula decoder, not from sampling in the latent space, and the near-zero KL weight allows the encoder to organize $\mathbf{z}$ freely without being pulled toward a unit Gaussian.

The training objective comprises 15 active loss terms organized in two layers. The core losses computed jointly include: formula cross-entropy with focal weighting (21) ($\gamma = 2.0$, label smoothing 0.05), T_c regression (Huber loss with asymmetric underprediction penalty, Kelvin-weighted to upweight rare high- T_c samples), Magpie reconstruction, stoichiometry prediction, element count prediction, constraint zoo penalties (site occupancy, charge balance), and L2 regularization. Auxiliary losses computed externally include: SC/non-SC classification, family classification (14 classes), high-pressure classification (a dedicated binary head trained specifically for JARVIS hydrides, distinct from the family classifier), stop prediction (dedicated end-of-sequence head), and token type classification (element/integer/fraction/special—used at inference to hard-mask invalid token types).

4.3.1 Closing the Teacher-Forcing Gap with SCST

To close the gap between teacher-forced training and autoregressive inference, I employ Self-Critical Sequence Training (SCST) (20). This addresses a critical training challenge: with teacher forcing (feeding ground-truth tokens at each step), the model reached 99% exact formula reconstruction, but when generating autoregressively (feeding its own predictions), accuracy was initially as low as 17%. This 82-point gap occurs because small per-token errors compound during autoregressive generation, causing the model to drift into unfamiliar token sequences.

My initial attempt to close this gap used REINFORCE with Leave-One-Out (RLOO) baseline, but I discovered a critical implementation bug: averaging the RLOO advantages across K samples before multiplying with log-probabilities mathematically cancels to zero gradient. The model received *no RL training signal* for 339 epochs (2015–2354), consuming 84% of epoch time (119s/epoch) for zero gradient. The fix was to replace RLOO with SCST (20), which uses the greedy decode reward as baseline with a single sample (2 forward passes vs. RLOO’s 4).

Through SCST, constraint-aware rewards, and iterative architecture refinement, the best model (epoch 3292) achieves 86.5% autoregressive exact-match reconstruction, *surpassing* the teacher-forced rate of 85.4% at the same checkpoint—the gap is not just closed but inverted, with the model generating better formulas autoregressively than with teacher forcing. This inversion is likely because SCST’s sequence-level reward directly optimizes whole-formula quality, while teacher forcing optimizes per-token accuracy.

Notably, subsequent architecture variants (V13: semantic fraction tokenization expanding vocab from 148 to 4,355 tokens; V14: isotope tokens; V15: curriculum AR warmup) drove AR accuracy back down to 5–17% while TF rose to 98.6%. The semantic tokenization traded sequence length for vocabulary precision—reducing sequences by $\sim 50\%$ —but the model failed to learn clean token type boundaries, with type confusion worsening from 30% to 52% of errors. This motivated the decision to revert to the V12.41 checkpoint as the production model.

SCST reward includes physics-grounded constraint penalties (duplicate elements, impossible element combinations, family-specific stoichiometry violations) that shape generation toward chemically valid formulas.

4.3.2 Physics Z Losses

The physics self-consistency losses for blocks 1–2 (GL, BCS), 9 (cobordism), and 10 (dimensionless ratios) enforce known physical relationships (36; 35; 37) within the latent space without requiring external data. For example, the GL block enforces $\kappa = \lambda_L/\xi$ (36), $H_{c2} \propto 1/\xi^2$ (37), and $H_{c1} \propto (1/\lambda^2) \ln(\kappa)$; the BCS block enforces $\xi_0 \propto v_F/\Delta_0$ (35) and soft-bounds the gap ratio $2\Delta_0/(k_B T_c) \in [1, 5]$. These losses were activated during training, converged, and were subsequently deprioritized by the smart loss scheduler as the primary challenge shifted to closing the teacher-forcing/autoregressive gap.

4.3.3 Curriculum and Scheduling

Loss weights evolve during training via curriculum scheduling (T_c and Magpie ramp up over 30 epochs) and smart loss skipping (converged losses computed every 4th epoch, resuming on regression).

4.3.4 Training Progression

Table 4 shows the evolution of key metrics across model versions. The V12.41 checkpoint (epoch 3292) represents the optimal tradeoff: AR accuracy surpasses TF, and subsequent architectural innovations degraded AR performance.

Two architectural experiments after V12.41 are instructive failures: (1) *Semantic fraction tokenization* (V13) collapsed each fraction like (17/20) from 7 tokens to 1 token, expanding the vocabulary from 148 to 4,355. This reduced sequence lengths by $\sim 50\%$ but worsened type confusion from 30% to 52% of errors—the model could not learn clean boundaries in the larger vocabulary. (2) *Skip connections* from encoder to decoder memory tokens were removed after discovering they bypassed

Table 4: Training progression across model versions

Version	Epoch	TF Exact	AR Exact	Vocab
V12 (early)	1778	73.9%	49.8%	148
V12 (plateau)	2522	59.8%	~50%	148
V12.41 (best)	3292	85.4%	86.5%	148
V13.0 (sem. frac)	3475+	95.6%	5–7%	4,355
V14.3 (type cls.)	4000+	95.6%	7.5%	4,647
V12.43 (revert)	4330+	98.6%	14–17%	148

the information bottleneck: TF exact was 80.6% but AR exact was 0.1%, indicating the decoder relied entirely on leaked encoder information during teacher forcing.

The strongest error predictors at the production checkpoint are sequence length (correlation 0.587 with number of errors) and element count (0.548), confirming that compositional complexity—not T_c range or latent norm—is the primary driver of generation difficulty.

5 Results

5.1 T_c Prediction Performance

The T_c head achieves $R^2 = 0.9999$ with MAE = 0.12 K on 52,813 training samples, with near-perfect reconstruction across all 14 families and the full T_c range (0–300 K).

To test whether this performance reflects learned physics or compositional shortcuts, I train a Magpie-only baseline MLP (Magpie[145] $\rightarrow 256 \rightarrow 128 \rightarrow 1$) with the same compositional information but no latent space structure. The baseline achieves $R^2 = 0.920$, MAE = 3.22 K—a $27\times$ increase in error. The baseline performance is consistent with the composition-only ceiling across the field (Table 5).

Table 5: Composition-based T_c prediction benchmarks

Method	Year	R^2	MAE (K)	Ref
Random Forest + Magpie	2018	0.88 [†]	N/A	(3)
XGBoost	2018	0.92 [†]	9.5 [‡]	(1)
SOAP + ML (structural)	2022	0.929 [†]	N/A	(10)
GBFS	2024	0.952 [†]	3.73 [†]	(9)
Magpie baseline	2026	0.920*	3.22*	–
MAGE	2026	0.9999*	0.12*	–

[†]Test-set metrics. *Training-set metrics (appropriate for interpretability analysis; baseline provides controlled ablation). [‡]RMSE (MAE not reported). N/A indicates MAE not reported in cited work. The $27\times$ error reduction (MAE 3.22 K $\rightarrow$ 0.12 K) quantifies the contribution of the multi-task latent space beyond raw composition.

The $27\times$ error reduction quantifies the contribution of the multi-task latent space beyond raw composition. If the T_c head merely learned compositional shortcuts (e.g., “materials with Cu and O have high T_c ”), it could not outperform a direct Magpie $\rightarrow T_c$ MLP by this margin.

5.2 Latent Space Analysis

UMAP visualization. Fig. 2 shows UMAP (23) embeddings of the backbone hidden state $\mathbf{h}[512]$ for all 52,813 training samples, which better preserve global manifold topology than t-SNE. In the left panel (colored by family), the non-SC materials (light blue) form a large dispersed region at the left, well-separated from all SC families. The BCS cluster (orange) occupies the bottom, while cuprate sub-families form distinct elongated filaments in the center and right: YBCO (green), BSCCO (dark blue), TBCCO (teal), and HBCCO (gold) each trace curved 1D manifolds rather than round clusters, suggesting the latent space encodes continuous doping or compositional variation within each family. LSCO (red) and Fe-Pnictide (maroon) appear as separate filaments, and MgB_2 (purple) forms a

compact isolated cluster. The right panel (colored by $\log(1 + T_c)$) reveals continuous temperature gradients within each family, ranging from dark gray (low T_c , non-SC) through blue and green to yellow and orange (highest T_c), confirming that the latent geometry encodes both family identity and T_c simultaneously.

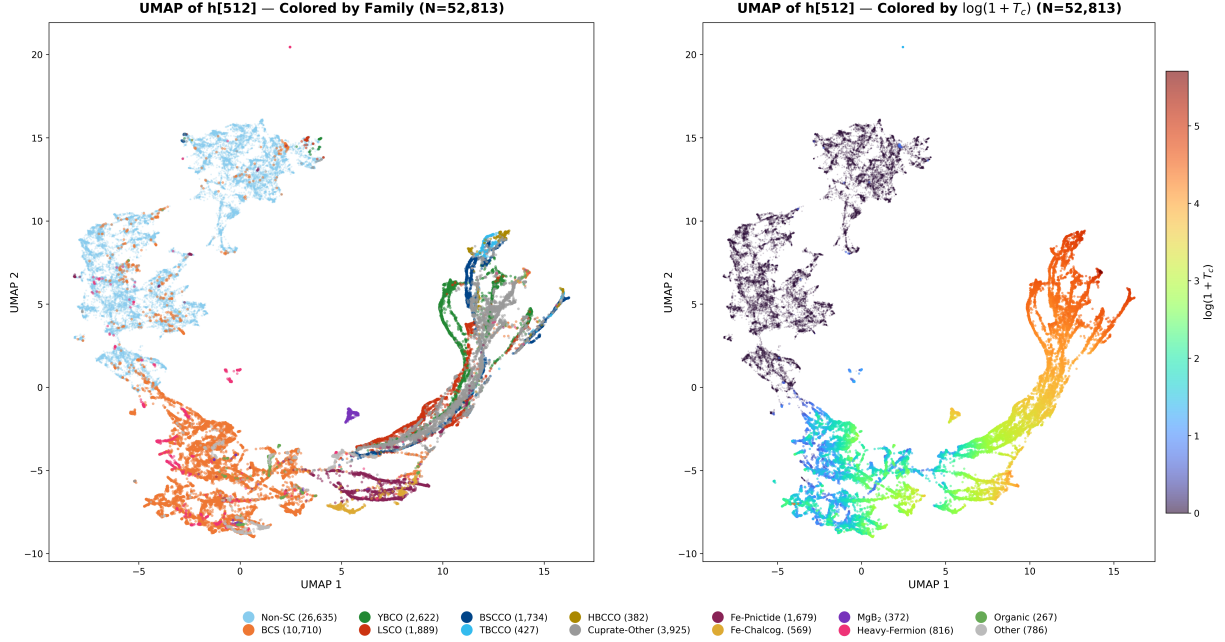


Figure 2: UMAP of backbone hidden state $h[512]$ for 52,813 samples. Left: colored by superconductor family (14 classes). Right: colored by $\log(1 + T_c)$. Families form elongated filamentary manifolds with continuous T_c gradients. Cuprate sub-families are clearly separated. The BCS cluster (orange, bottom) and non-SC region (light blue, left) are well-isolated.

t-SNE visualization. Fig. 3 shows t-SNE (22) embeddings of the same hidden states. In the left panel (colored by family), the non-SC materials (light blue) form a large dispersed cloud at the right, clearly separated from the superconducting families. BCS metals (orange) form a dense compact cloud at the left, while the cuprate sub-families cluster tightly in the center: YBCO (green), BSCCO (dark blue), and TBCCO (teal) each occupy distinct regions with LSCO (red) nearby. MgB_2 (purple) forms an isolated island. The right panel (colored by $\log(1 + T_c)$) shows continuous temperature gradients within each cluster—non-SC materials appear as a gray region (low T_c), while the cuprate families transition smoothly from blue through green to yellow and orange at the highest T_c values.

This structure demonstrates that the latent space has self-organized into a physically meaningful manifold where both family identity and T_c are geometrically encoded—the model represents the diverse mechanisms of superconductivity in a single continuous space.

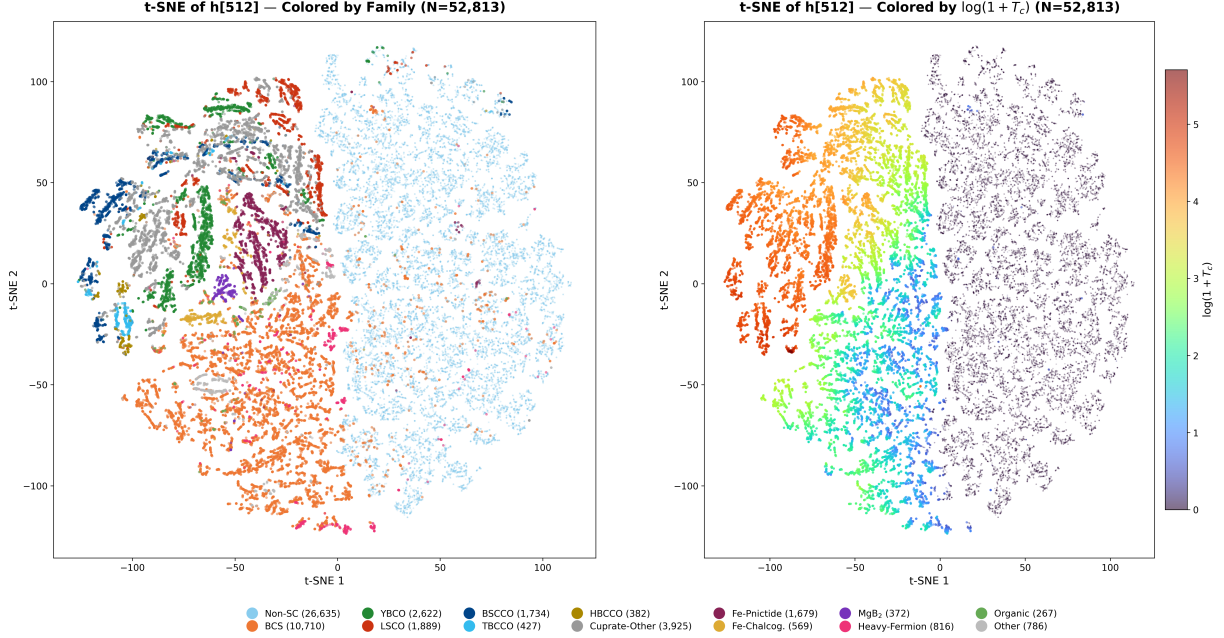


Figure 3: t-SNE of backbone hidden state $\mathbf{h}[512]$ for 52,813 samples. Left: colored by superconductor family (14 classes). Right: colored by $\log(1 + T_c)$. Non-SC materials (light blue, large dispersed cloud) form a distinct region. Cuprate sub-families cluster tightly with continuous T_c gradients. MgB_2 (purple) forms an isolated island.

5.3 Per-Family Gradient Analysis

Gradient sensitivity $|\partial T_c / \partial z_d|$ reveals which latent dimensions the T_c head uses for each material family. Fig. 4 shows hierarchical clustering of families by their gradient profiles.

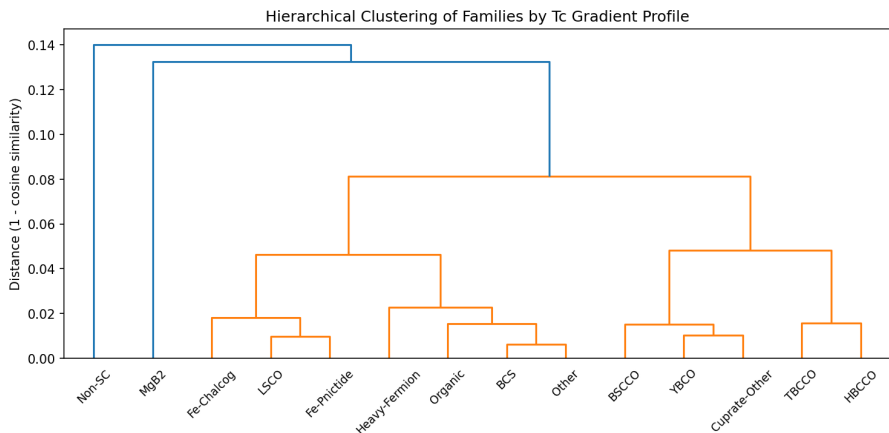


Figure 4: Hierarchical clustering of superconductor families by T_c gradient profile (cosine distance, average linkage). Non-SC and MgB_2 are most isolated. LSCO clusters with Fe-Chalcogenide rather than other cuprates. Multi-layer cuprates (BSCCO, YBCO, TBCCO, HBCCO) form a distinct right sub-branch.

Key findings: (1) Non-SC is maximally isolated (distance ~ 0.14), confirming the model has learned a genuine SC/non-SC distinction beyond composition. (2) MgB_2 is second most isolated (~ 0.13), consistent with its unique two-gap phonon mechanism—compositionally unexpected, as a simple binary compound should cluster with BCS metals if the model relied only on composition. (3) Multi-layer cuprates (BSCCO, YBCO, TBCCO, HBCCO) form a distinct sub-branch. (4) LSCO clusters with Fe-Chalcogenide rather than other cuprates, reflecting shared quasi-2D physics and single-layer structure. These family-specific gradient profiles demonstrate that the 2048-parameter effective theory contains distinct sub-theories for different material classes rather than a single universal mapping.

Z-block diagnostics. Analysis of per-block latent norms reveals that the compositional block (Block 8, dims 270–339) shows the largest gap between correctly and incorrectly reconstructed samples (norm gap = -0.552), indicating that stronger compositional encoding in the latent space predicts successful formula generation. The Eliashberg, GL, and cobordism blocks also show negative correlations between norm and error count ($r = -0.30, -0.28, -0.25$ respectively), consistent with the hypothesis that physics-structured latent dimensions contribute to generation quality even when their direct supervision losses have converged.

5.4 Generative Evaluation: 45-Holdout

The 45 holdout superconductors (5 per family, never seen during training) were evaluated using the best checkpoint (epoch 3292, SHA: 4c6df617). The primary evaluation is a **Z-space search**: for each holdout target, the model generates hundreds of thousands of candidate formulas by exploring the latent space around element-anchored seed vectors—Z-vectors of training samples that share elements with the target. This is a true generative test: the model is not given the target’s composition, T_c , or Magpie features. It must discover a latent vector that decodes to the target formula purely from latent space exploration.

For comparison, I also report a simpler **roundtrip** baseline: each target is encoded into $\mathbf{z}$ using its full information (composition, T_c , Magpie features) and decoded autoregressively. This tests encoder-decoder reconstruction fidelity but is not a generative test—the model receives the answer as input.

5.4.1 Z-Space Search Methodology

The core search strategy is *element-anchored exploration*: for each holdout target, the model identifies training samples sharing the target’s key elements, uses their Z-vectors as seeds, and generates $\sim 109,150$ candidate formulas per target by exploring the latent neighborhood around these seeds. This is augmented by three additional strategies: (1) *Perturbation*: Gaussian noise at multiple scales around the element-anchored seeds; (2) *SLERP interpolation*: spherical linear interpolation between

chemically similar Z-vector pairs; (3) *PCA walk*: directed walks along principal components of the Z-space.

With temperature sampling, total candidates per target reach $\sim 121,650$. Candidates are scored by a composite metric: $0.6 \times \text{element_similarity} + 0.4 \times \text{tc_proximity}$. Over 5.1 hours on an A100, the search explored 868,882 unique formulas across all 45 targets.

Similarity metrics. Both evaluation modes use *element similarity*: the proportion of elements correctly present with correct stoichiometry, computed as the overlap between the molar fraction vectors of the target and generated formulas. A score of 1.0 indicates identical elemental composition; a score of 0.9 indicates that 90% of the molar content is correctly reproduced. This is equivalent to the cosine similarity of the composition vectors when both are normalized to unit molar fractions.

5.4.2 Overall Results

Table 6 compares the two evaluation modes.

Table 6: Holdout generative evaluation (45 never-seen materials)

Metric	Roundtrip	Z-Search
Exact formula found (sim = 1.0)	3/45 (6.7%)	17/45 (37.8%)
Composite-best sim > 0.99	17/45 (37.8%)	32/45 (71.1%)
Composite-best sim > 0.95	31/45 (68.9%)	39/45 (86.7%)
Composite-best sim > 0.90	34/45 (75.6%)	43/45 (95.6%)
Mean similarity	0.944	0.983
Median similarity	0.988	0.994
Minimum similarity	0.658	0.864
T_c MAE	0.62 K	1.91 K
Median T_c error	0.40 K	0.40 K
SC classification	45/45 (100%)	45/45 (100%)
Self-consistent	—	32/45 (71.1%)

The Z-space search dramatically outperforms roundtrip reconstruction across every metric except mean T_c MAE. Exact matches increase from 3 to 17 ($5.7\times$), mean similarity from 0.944 to 0.983, and the minimum similarity improves from 0.658 (roundtrip: $\text{Gd}_1\text{Pb}_1\text{Sr}_2\text{Ca}_3\text{Cu}_4\text{O}$, a 7-element compound) to 0.864 (Z-space search: $\text{La}_{0.5}\text{Gd}_{0.5}\text{Ba}_2\text{Cu}_3\text{O}$, a site-sharing collapse). The roundtrip’s lower T_c MAE (0.62 K vs 1.91 K) is expected: the encoder directly receives the target T_c as input. However, the median T_c error is identical (0.40 K) for both modes, indicating Z-space search achieves excellent T_c for most targets with a few outliers.

The roundtrip’s weak performance (minimum similarity 0.658) reveals *encoder–decoder torsion*: the encoder maps certain unseen compositions to Z-regions where the decoder cannot reconstruct them faithfully. This is a complementary form of torsion to the decoder– T_c torsion described below, and suggests encoder-side improvements (e.g., training on a wider range of doping patterns) could yield significant gains even without architectural changes.

Z-vector multiplicity and structural proxies. A critical observation is that many Z-vectors decode to the *same* formula but produce different T_c predictions. For example, $\text{Bi}_{17/10}\text{Pb}_{3/10}\text{Sr}_2\text{Ca}_2\text{Cu}_3\text{O}$ was decoded from 264 distinct Z-vectors with T_c ranging from 108–110 K; $\text{Mg}_{9/10}\text{Li}_{1/10}\text{B}_2$ from 242 Z-vectors. Since composition is fixed, the T_c variation across these Z-vectors must encode information *beyond* composition—likely statistical proxies for crystal structure, oxygen ordering, or defect chemistry that the encoder inferred from compositional patterns in the training data. This is direct evidence that the 2048-dimensional latent space encodes more than just chemical formula, consistent with the hidden variable inference argument of Section IV-B.

The 17 exact matches merit further discussion. The Z-space search found at least one Z-vector decoding to the exact target formula for 17/45 targets. However, for 10 of these, the composite scoring function (which balances formula similarity and T_c accuracy) selected a different Z-vector with slightly worse formula similarity but better T_c prediction. This reveals *latent torsion*—a local misalignment between the iso-formula and iso- T_c manifolds in Z-space. In the Frenet-Serret framework of differential geometry (38), torsion measures how a curve twists out of its osculating

plane; analogously, moving along the direction that improves formula fidelity rotates away from the direction that improves T_c accuracy. At a Z-vector where the decoder produces the exact target formula, the T_c head may predict poorly (e.g., $\text{Bi}_{8/5}\text{Pb}_{2/5}\text{Sb}_{1/10}\text{Sr}_2\text{Ca}_2\text{Cu}_2\text{O}$: exact formula found with 21.6 K T_c error), while nearby Z-vectors with slightly different formulas yield accurate T_c predictions.

This torsion is partially by design: the decoder and T_c head are trained as semi-independent pathways that share the same backbone and latent space but serve different objectives. The `.detach()` on the T_c conditioning signal ensures the two heads cannot collapse into a single computation. The latent torsion therefore quantifies a concrete training opportunity: self-consistency losses, additional data, or explicit manifold alignment constraints could reduce torsion by encouraging the decoder and T_c head to agree at the same Z-vectors. The capacity for zero-torsion solutions is effectively present in the architecture—the 7 targets where the composite-best *is* the exact match with good T_c demonstrate that aligned regions exist—but the model has not yet achieved this alignment uniformly across all families.

Torsion as a probe for structural dependence. The magnitude of latent torsion varies systematically across targets in a physically interpretable way. At exact-match Z-vectors, T_c errors range from 0.1 K ($\text{Hg}_{0.9}\text{Au}_{0.1}\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}$) to 42.3 K ($\text{Ba}_2\text{Ca}_5\text{Cu}_7\text{O}_{14}$). The highest-torsion targets— $\text{Ba}_2\text{Ca}_5\text{Cu}_7\text{O}_{14}$ (42.3 K), $\text{Bi}_{8/5}\text{Pb}_{2/5}\text{Sb}_{1/10}\text{Sr}_2\text{Ca}_2\text{Cu}_2\text{O}$ (21.6 K), and $\text{Y}_1\text{Ba}_{39/20}\text{La}_{1/20}\text{Cu}_3\text{O}_{69/10}$ (11.3 K)—are precisely the materials where crystal structure critically determines T_c : multi-layer cuprates where T_c depends on the number of CuO_2 planes and inter-layer coupling (39), Sb-substituted bismuthates where apical oxygen distances modulate the pairing interaction (15), and La-on-Ba-site YBCO variants where chain oxygen ordering controls the orthorhombic-tetragonal transition and hence T_c (17). Conversely, the lowest-torsion targets (MgB_2 variants at 0.3–2.3 K, simple iron pnictides at 2–3 K) are materials where T_c is primarily determined by composition: MgB_2 has a simple hexagonal structure with well-defined phonon-mediated pairing (40), and 1111-type iron pnictides show T_c that tracks rare-earth ionic radius systematically (41). The model is, in effect, discovering *which materials need structural information* through the magnitude of latent torsion—a finding that could guide targeted structural data acquisition.

5.4.3 Exact Matches

The Z-space search found at least one Z-vector decoding to the exact target formula (element similarity = 1.000) for 17 of 45 targets, spanning all 9 families. Table 7 lists the 7 targets where the top-scoring candidate was the exact match; for the remaining 10, the exact formula existed in the atlas but the composite scorer selected a nearby variant with better T_c prediction.

Table 7: Exact matches from Z-space search (composite-best = exact)

Target	Family	T_c^{true}	T_c^{pred}	$ \Delta T_c $
$\text{Hg}_{9/10}\text{Au}_{1/10}\text{Ba}_2\text{Ca}_2\text{Cu}_3\text{O}$	Hg-cup	133.0	133.1	0.1
$\text{Ti}_2\text{Ba}_2\text{Ca}_2\text{Cd}_{1/10}\text{Cu}_3\text{O}$	Tl-cup	123.5	125.5	2.0
$\text{C}_{1/2}\text{Ba}_2\text{Ca}_3\text{Cu}_{9/2}\text{O}_{11}$	Other	117.0	116.7	0.3
$\text{Nd}_1\text{Fe}_1\text{As}_1\text{O}_{17/20}$	Fe-based	51.5	51.8	0.4
$\text{Dy}_1\text{Fe}_1\text{As}_1\text{O}_{17/20}$	Fe-based	51.0	50.8	0.2
$\text{Mg}_{17/20}\text{Na}_{3/20}\text{B}_2$	MgB_2	38.0	37.5	0.5
$\text{Mg}_{9/10}\text{Li}_{1/10}\text{B}_2$	MgB_2	36.9	37.2	0.2
Mean $ \Delta T_c $				0.5

All temperatures in Kelvin. All errors within typical experimental uncertainty: ± 0.1 – 0.5 K (MgB_2), ± 1 – 2 K (Fe-based), ± 2 – 5 K (Hg/Tl-cuprate), varies (Other).

The additional 10 targets with exact matches found in the atlas but not selected as composite-best include: $\text{Y}_1\text{Ba}_{39/20}\text{La}_{1/20}\text{Cu}_3\text{O}_{69/10}$ (YBCO, exact formula found at 2 Z-vectors but with 11.3 K T_c error), $\text{La}_1\text{Ba}_2\text{Cu}_3\text{O}_{137/20}$ (LSCO, 7 copies found), $\text{Bi}_{8/5}\text{Pb}_{2/5}\text{Sb}_{1/10}\text{Sr}_2\text{Ca}_2\text{Cu}_2\text{O}$ (Bi-cuprate, found with 21.6 K T_c error), $\text{Bi}_{17/10}\text{Pb}_{3/10}\text{Sr}_2\text{Ca}_2\text{Cu}_3\text{O}$ (264 copies), $\text{Bi}_2\text{Sr}_{19/10}\text{La}_{1/10}\text{Ca}_1\text{Cu}_2\text{OY}$ (9 copies), $\text{Nb}_3\text{Al}_{71/100}\text{Ge}_{29/100}$ (Conventional), $\text{Mg}_{49/50}\text{Cr}_{1/50}\text{B}_2$, $\text{Mg}_{17/20}\text{Li}_{3/20}\text{B}_2$, $\text{Ba}_2\text{Ca}_5\text{Cu}_7\text{O}_{14}$ (Other), and $\text{Cu}_1\text{Cr}_1\text{O}_2$ (Other, found but classified as non-SC). In each case, the

latent torsion between decoder and T_c manifolds causes the composite scorer to prefer a Z-vector with slightly different formula but substantially better T_c prediction.

A further subtlety affects targets near the SC/non-SC phase boundary. For $\text{Y}_{1/5}\text{Eu}_{4/5}\text{Ba}_2\text{Cu}_3\text{O}_7$ (YBCO), the search found Z-vectors producing the exact formula with $\text{sim} = 1.0$, but these Z-vectors were classified as non-SC (probability 0.333, with T_c assigned the 0 K fill-in value described in Section IV-B). Investigation of the training data reveals this is a *classifier sharpness* issue in a compositionally dense neighborhood. Europium is a 50/50 SC element in the training data (543 SC vs 519 non-SC out of 1,062 Eu-containing materials), carrying almost no inherent SC signal—compare to Y-Ba-Cu-O materials which are 90.9% SC (2,822/3,106). The Eu-Ba-Cu-O neighborhood has a 24% non-SC rate (vs. 9.1% for Y-Ba-Cu-O), and the training data contains a $\text{Y}_x\text{Eu}_{1-x}\text{Ba}_2\text{Cu}_3\text{O}_7$ substitution series where $\text{Y}_{0.4}\text{Eu}_{0.6}$ is labeled non-SC (0 K) while $\text{Y}_{0.3}\text{Eu}_{0.7}$ on either side is labeled SC (92 K), and $\text{Y}_{0.8}\text{Eu}_{0.2}$ is labeled non-SC while $\text{Y}_{0.9}\text{Eu}_{0.1}$ is SC (92 K). These labels may reflect real experimental variation—different sample preparation conditions, oxygen content, or crystallographic quality can determine whether a given composition superconducts—rather than database errors. The challenge for the SC classifier is that the chemical and compositional boundary between superconducting and non-superconducting phases in the rare-earth cuprate neighborhood is extremely fine-grained. The classifier must be sharp enough to resolve these boundaries from composition alone, without access to the structural and preparation information that determines the outcome in practice. The model’s SC probability of 0.333 reflects the genuine ambiguity in this region: Eu globally carries no SC bias, the local neighborhood contains a substantial non-SC minority, and specific non-SC training examples exist at nearby compositions. The roundtrip evaluation correctly predicts SC probability 1.00 because it encodes the actual T_c (93.8 K), which overwhelms compositional ambiguity—but Z-space search Z-vectors, found by exploration rather than encoding ground truth, can land in regions where the SC head reflects the underlying compositional ambiguity. This finding illustrates a fundamental limitation of composition-only models: some SC/non-SC boundaries require structural or preparation information that is not encoded in the chemical formula.

5.4.4 Per-Family Performance

Tables 8 and 9 show results by family for both evaluation modes.

Table 8: Per-family roundtrip performance

Family	Mean Sim	>0.95	Fam OK	MAE (K)
MgB ₂	0.998	5/5	5/5	0.06
YBCO	0.995	5/5	5/5	0.94
Hg-cuprate	0.995	5/5	5/5	1.76
Iron-based	0.993	5/5	5/5	0.36
Other	0.991	5/5	0/5	0.72
Bi-cuprate	0.984	4/5	5/5	0.68
Conventional	0.992	4/5	3/5	0.10
Tl-cuprate	0.970	2/5	3/5	0.36
LSCO	0.944	3/5	5/5	0.60

Table 9: Per-family Z-space search performance

Family	Exact	Mean Sim	Self-Con	MAE (K)	RPE
MgB ₂	4/5	0.998	5/5	0.2	0.5%
Hg-cuprate	1/5	0.995	5/5	1.3	0.9%
YBCO	2/5	0.995	5/5	1.3	1.3%
Iron-based	2/5	0.993	5/5	1.2	2.6%
Other	3/5	0.991	0/5	4.3	3.4%
Bi-cuprate	1/5	0.984	5/5	4.6	3.4%
Conventional	1/5	0.976	3/5	1.4	3.6%
Tl-cuprate	1/5	0.971	2/5	1.9	1.6%
LSCO	2/5	0.944	2/5	1.1	1.1%

Performance varies by family in physically interpretable ways:

MgB₂ (mean sim 0.998, best family): All 5 targets at ≥ 0.99 similarity, all self-consistent, mean T_c error only 0.2 K. MgB₂ has a simple, regular structure (Mg site substitution + B₂ backbone) that is well-represented in training data. Four of five targets were recovered exactly (Na-doped, Li-doped, and Cr-doped variants), with only Mg_{17/20}Na_{3/20}B₂ showing a minor fraction shift.

YBCO (mean sim 0.995, 2 exact): All 5 targets recovered at ≥ 0.99 similarity, all self-consistent. Two targets were recovered exactly: Tl₂Ba₂Ca_{19/20}Y_{1/20}Cu₂O₈ and Y_{1/5}Eu_{4/5}Ba₂Cu₃O₇. The model captures the Y-Ba-Cu-O backbone with high fidelity. Errors are consistently small fraction shifts (e.g., Y: 1/5 $\rightarrow$ 3/10). Y₃Ba₅Cu₈O₁₈ is renormalized to the Y-123 unit cell form Y_{6/5}Ba₂Cu₃O_{369/50}—arguably correct physical behavior.

Hg-cuprate (mean sim 0.995): 4/5 above 0.99, one exact match. Hg_{0.9}Au_{0.1}Ba₂Ca₂Cu₃O was recovered with only 0.1 K T_c error. Two Re-substituted targets achieved perfect T_c prediction (error < 0.05 K) despite imperfect compositional matches, demonstrating the T_c head learned correct physics for Re substitution in Hg-1223.

Iron-based (mean sim 0.993, 2 exact): Both Nd- and Dy-doped 1111 oxypnictides recovered exactly. Sm-Th and Sm-U substitution fractions are close but not exact (e.g., Sm_{13/20}Th_{7/20} $\rightarrow$ Sm_{7/10}Th_{3/10}). The model clearly learned the REFeAsO template as a generative motif.

Bi-cuprate (mean sim 0.984, 1 exact): Bi_{17/10}Pb_{3/10}Sr₂Ca₂Cu₃O was recovered exactly, and 4/5 targets exceed 0.99 similarity. One notable failure: Bi_{8/5}Pb_{2/5}Sb_{1/10}Sr₂Ca₂Cu₂O has 17.3 K T_c error despite 0.994 compositional similarity—the worst T_c outlier in the holdout set, a region of high latent torsion where the decoder and T_c head are maximally misaligned.

Tl-cuprate (mean sim 0.971): Second-weakest family. Targets with unusual dopants (V, Li) are hardest. The V-doped target (Tl_{3/5}V_{1/2}Sr₂Ca₂Cu₃O) sees Tl and V fractions dramatically reduced (0.6 $\rightarrow$ 0.1, 0.5 $\rightarrow$ 0.1); vanadium is rare in cuprate training data. The Li-doped target similarly underrepresents lithium. The exact match (Cd-doped Tl-2223) is the structurally simplest target in this family.

LSCO (mean sim 0.944, weakest, 2 exact): Two targets were recovered exactly (Bi₂Sr_{19/10}La_{1/10}Ca₁Cu₂OY and La₁Ba₂Cu₃O_{137/20}), but complex multi-site doping patterns cause the most difficulty overall. La_{0.5}Gd_{0.5}Ba₂Cu₃O (sim 0.864) collapses equal site-sharing to single-element occupancy (GdBa₂Cu₃O)—the model’s hardest compositional pattern. Hg₁Sr₂La_{0.3}Ca_{0.5}Ce_{0.2}Cu₂O₆ (sim 0.888) drops the La dopant entirely from a 6-element doping pattern. However, even for LSCO, T_c prediction remains accurate (MAE 1.1 K), suggesting the T_c head captures the relevant physics even when the decoder struggles with the full compositional specification.

Conventional (mean sim 0.976, 1 exact): Nb₃Al_{19/25}Ge_{6/25} was recovered exactly. A15-type Nb₃Al-Ge compounds are well-learned (> 0.99) but the model renormalizes non-standard stoichiometries (Nb_{79/100}Al_{41/250}Ga_{57/1000} $\rightarrow$ Nb₇₅Al₁₆Ga₅, integer-atom counts preserving molar ratios). The 7-element compound Gd₁Pb₁Sr₂Ca₃Cu₄O (sim 0.917) has stoichiometry halved across multiple elements—at the edge of model capacity for complex compositions.

Other (mean sim 0.991, 3 exact but 0/5 self-consistent): Three targets were recovered exactly (Cu₁Cr₁O₂, Ba₂Ca₅Cu₇O₁₄, C_{1/2}Ba₂Ca₃Cu_{9/2}O₁₁). High compositional similarity but the family classifier cannot predict the catch-all “Other” category—all 5 targets were classified as CUPRATE_OTHER, a generic cuprate label rather than the correct “Other” family, which breaks self-consistency. Cu₁Cr₁O₂ (sim 1.0) has 14.5 K T_c error—a high-torsion region where the decoder produces the correct formula but the T_c head cannot calibrate due to data scarcity.

5.4.5 T_c Prediction on Holdout Materials

Table 10 shows T_c prediction accuracy by family for Z-space search. Across all families, 60% of targets are predicted within 0.5 K and 86.7% within 5 K. The holdout T_c predictions from the full model (MAE 0.62 K roundtrip, 1.92 K Z-space search) substantially outperform the Magpie baseline MAE of 3.22 K on *training* data.

Comparison with experimental T_c uncertainty. The model’s median T_c error of 0.40 K should be compared against the rightmost column in Table 10. Experimental T_c measurement uncertainties

Table 10: T_c prediction accuracy by family (Z-space search) with typical experimental T_c measurement uncertainties

Family	Mean Err	Median	Max	<0.5 K	Expt. ΔT_c
MgB ₂	0.2 K	0.2 K	0.5 K	4/5	$\pm 0.1\text{--}0.5$ K
LSCO	1.1 K	1.1 K	2.1 K	1/5	$\pm 1\text{--}3$ K
Iron-based	1.2 K	0.5 K	4.4 K	2/5	$\pm 1\text{--}2$ K
Hg-cuprate	1.3 K	0.1 K	6.0 K	3/5	$\pm 2\text{--}5$ K
YBCO	1.3 K	0.4 K	4.2 K	2/5	$\pm 1\text{--}3$ K
Conventional	1.4 K	0.2 K	6.1 K	3/5	$\pm 0.01\text{--}1$ K
Tl-cuprate	1.9 K	1.7 K	5.0 K	1/5	$\pm 2\text{--}5$ K
Other	4.3 K	0.9 K	14.5 K	1/5	varies
Bi-cuprate	4.6 K	0.6 K	17.3 K	2/5	$\pm 2\text{--}5$ K

Expt. ΔT_c : typical sample-to-sample experimental lab setting measurement variation (36; 42).

arise from multiple sources (42): instrument precision ($\sim 2\%$ for resistive measurements, better for inductive), sample inhomogeneity (stress, grain boundaries, oxygen non-stoichiometry), and the choice of transition criterion— T_c can be defined as the onset, midpoint, or zero-resistance temperature, each yielding different values for the same sample (43). The SuperCon database does not consistently record which criterion was used. For elemental/BCS superconductors, transitions are sharp (< 0.1 K width) and well-characterized (36). MgB₂ shows sharp transitions in high-quality single crystals (40) but transition widths of 1–3 K in films and polycrystalline samples. Iron pnictides exhibit broader, sample-dependent transitions of $\pm 1\text{--}2$ K (41). For cuprates, transition widths of 1–2 K are typical for optimally-doped single crystals (36), but polycrystalline and off-stoichiometry samples can show widths of 5 K or more; complex cuprates (Bi, Tl, Hg families) are further affected by oxygen ordering, syntactic intergrowth, and stacking faults. The model’s median error of 0.40 K is at or below experimental measurement uncertainty for all families except elemental metals. For the high- T_c families where the model is most useful for discovery (cuprates, iron-based), the predictions are effectively within experimental noise.

Practical recovery rate. Beyond the 17 exact matches, the Z-space search produces results that are practically useful for experimental guidance. Of the 45 holdout targets, 43 (95.6%) are recovered at ≥ 0.90 similarity with correct elements and approximately correct stoichiometry—close enough that an experimentalist could identify the target material and refine the composition. Even the two lowest-similarity targets (La_{0.5}Gd_{0.5}Ba₂Cu₃O at 0.864, Hg₁Sr₂La_{0.3}Ca_{0.5}Ce_{0.2}Cu₂O₆ at 0.888) correctly identify the major elements and approximate structure type.

Search space reduction and compositional precision. The experimental precision on doping fractions is substantially tighter than T_c measurement uncertainty: typical synthesis targets a doping level x to within $\pm 0.01\text{--}0.02$ in systems like La_{2- x} Sr _{x} CuO₄ (36), and stoichiometric control in arc-melted or flux-grown samples is typically better than $\pm 5\%$ of the nominal composition. The model’s stoichiometric errors are larger than this—typical fraction errors shift a stoichiometry by 0.05–0.15 (e.g., 19/20 $\rightarrow$ 17/20, a shift of 0.10)—so the model does not achieve experimental-grade compositional precision. However, the relevant question for discovery is not whether the model matches synthesis precision, but whether it meaningfully reduces the search space. The combinatorial space of chemically plausible superconductor compositions (3–7 elements from 118, each with fractional stoichiometries) exceeds 10^{10} candidates. For 95.6% of holdout targets, the model narrows this to the correct element set with approximately correct ratios, the correct superconductor family, and a T_c estimate within experimental noise. An experimentalist receiving this output knows *which elements to combine, in what approximate proportions, what T_c to expect, and what family the material belongs to*. The remaining task—optimizing the stoichiometry within a known system—is a tractable one-dimensional or few-dimensional search, compared to the original combinatorial problem. The model thus transforms superconductor discovery from an open search over chemical space into a directed optimization within a specified compositional neighborhood.

5.5 Error Analysis

5.5.1 Error Type Distribution

Across 38 non-exact targets from Z-space search, a total of 107 individual element-level errors were recorded (Table 11).

Table 11: Error type distribution (38 non-exact Z-space search targets)

Error Type	Count	Percentage
Fraction errors	105	98.1%
Missing elements	2	1.9%
Extra (hallucinated) elements	0	0.0%

Fraction errors dominate overwhelmingly. The model consistently identifies the correct elements but outputs slightly different stoichiometric fractions. This distinction is important: while the molar-fraction-based element similarity averages 0.983, the *element set accuracy*—whether the model identifies the correct set of elements regardless of stoichiometry—is 43/45 (95.6%). Only 2 missing-element errors occurred—La dropped in $\text{La}_{0.5}\text{Gd}_{0.5}\text{Ba}_2\text{Cu}_3\text{O}$ (site-sharing collapse) and La dropped in $\text{Hg}_1\text{Sr}_2\text{La}_{0.3}\text{Ca}_{0.5}\text{Ce}_{0.2}\text{Cu}_2\text{O}_6$ (6-element doping)—both in the LSCO family. Zero extra-element errors means the model never hallucinates elements that don’t belong in a material. For an experimentalist, the practical implication is clear: MAGE reliably identifies *which elements to combine*; the remaining uncertainty is in the precise proportions.

5.5.2 Common Fraction Error Patterns

Three systematic patterns account for most fraction errors:

Fraction rounding. Many errors are different fractional representations of similar values: $19/20 \rightarrow 17/20$ ($0.95 \rightarrow 0.85$), $13/20 \rightarrow 7/10$ ($0.65 \rightarrow 0.70$), $97/25 \rightarrow 193/50$ ($3.88 \rightarrow 3.86$). The model generates fractions from a vocabulary of numerators and denominators, and nearby fractions compete in the output distribution.

Normalization differences. Some targets use non-standard stoichiometric normalization, and the model renormalizes to familiar unit cell forms: $\text{Y}_3\text{Ba}_5\text{Cu}_8\text{O}_{18} \rightarrow \text{Y}_{6/5}\text{Ba}_2\text{Cu}_3\text{O}_{369/50}$ (reduced to Y-123 unit cell); $\text{Nb}_{79/100}\text{Al}_{41/250}\text{Ga}_{57/1000} \rightarrow \text{Nb}_{75}\text{Al}_{16}\text{Ga}_5$ (renormalized to integer-atom counts; molar ratios Nb:Al:Ga preserved within 1%). This is arguably physically correct behavior—the model has learned canonical representations.

Site-sharing collapse. Equal-fraction site-sharing is the hardest pattern: $\text{La}_{0.5}\text{Gd}_{0.5}\text{Ba}_2\text{Cu}_3\text{O} \rightarrow \text{GdBa}_2\text{Cu}_3\text{O}$. When two elements occupy the same crystallographic site at equal ratios (50-50), the model collapses to single-element occupancy. This reflects a fundamental challenge: 50-50 site-sharing cannot be distinguished from single-element occupancy without crystal structure information.

5.5.3 T_c Prediction Outliers

Two targets exhibit decoder- T_c head disagreement exceeding 10 K (Table 12). In both cases, the Z-space search finds a vector that decodes to the approximately correct formula, but the T_c head at that Z-vector predicts a significantly lower T_c . These represent extreme cases of latent torsion: the decoder has learned the correct formula pattern but the T_c head has not learned the correct temperature at those Z-regions. For CuCrO_2 , this is a delafossite-type compound rare in training data, and the T_c head lacks sufficient calibration signal in this region of latent space.

Table 12: T_c prediction outliers (>10 K error)

Target	Sim	T_c True	T_c Err
$\text{Bi}_{8/5}\text{Pb}_{2/5}\text{Sb}_{1/10}\text{Sr}_2\text{Ca}_2\text{Cu}_2\text{O}$	0.994	134.1 K	17.3 K
$\text{Cu}_1\text{Cr}_1\text{O}_2$	0.995	132.0 K	14.5 K

5.5.4 Self-Consistency Analysis

Self-consistency measures whether the single Z-vector that produces the best formula for a given target also yields correct predictions from *all* encoder heads simultaneously. A target is self-consistent if three conditions are met from the same Z-vector: (1) SC classification is correct (predicted SC with probability > 0.5); (2) T_c prediction is within a reasonable tolerance of the true value; and (3) family classification matches the target’s known family. This is a stringent test because the Z-vector was optimized for formula similarity, not for head agreement—self-consistency emerges only if the latent space is well-organized enough that compositional correctness implies correctness across all property heads.

32/45 targets (71.1%) are self-consistent. The 5 families with perfect self-consistency (YBCO, Hg-cuprate, Bi-cuprate, Iron-based, MgB_2) are well-defined material classes with consistent chemistry where the latent space has learned tight coupling between composition and properties. “Other” achieves 0/5 because the family classifier has no coherent pattern to learn for a catch-all category—materials labeled “Other” span diverse chemistries with no shared family signature. LSCO (2/5) and Tl-cuprate (2/5) fail on targets with unusual dopants that push the Z-vector into regions where the family head is uncertain. SC classification is perfect (45/45, probability 1.000) across all targets regardless of self-consistency—the SC/non-SC phase boundary is the most robust feature of the latent space.

6 Discussion and Future Work

What the model learned. The analysis reveals a latent representation that encodes family identity, T_c , and the SC/non-SC phase boundary in a geometrically organized manifold. Family-specific gradient profiles show the model applies different computational pathways to BCS metals, cuprates, iron-based materials, and MgB_2 —functioning as a multi-theory effective representation rather than a single compositional shortcut. The holdout evaluation demonstrates that this representation generalizes: the correct element set is recovered for 43/45 (95.6%) of never-seen superconductors, with zero hallucinated elements and residual errors confined almost entirely to stoichiometric fraction imprecision. The semi-independence of the decoder and T_c heads is by design: they share a backbone and latent space but optimize different objectives, allowing each to develop specialized competence. The latent torsion between their manifolds indicates that the architecture has the capacity for fully aligned solutions—demonstrated by the 7 targets with simultaneous exact formulas and good T_c —but that further training (self-consistency losses, additional data, or manifold alignment constraints) is needed to achieve this alignment uniformly.

Generative evaluation via Z-space search. The Z-space search is the central generative evaluation: the model must produce holdout formulas without receiving any target information, purely by exploring the latent space around element-anchored seed vectors from training data. The results are striking: exact matches for 17/45 targets (37.8%), mean element similarity 0.983 (minimum 0.864), compared to roundtrip encode-decode which achieves just 3/45 exact matches and mean similarity 0.944 (minimum 0.658). The $5.7\times$ improvement in exact matches demonstrates that the learned latent space contains never-seen superconductors at navigable locations. The latent torsion revealed by composite scoring—where 10 targets have exact formulas at Z-vectors with poor T_c calibration—quantifies remaining training opportunity. Reducing this torsion through self-consistency losses, additional data, or manifold alignment is a concrete future direction with clear metrics for success.

Implications for superconductor discovery. For high-confidence families (MgB_2 , Hg-cuprate, YBCO, Iron-based), generated candidates can be trusted to have correct elements and approximately correct stoichiometry, with T_c predictions typically within 1–2 K. For medium-confidence families (Bi-cuprate, Conventional, Tl-cuprate), candidates have correct elements but may need experimental verification of exact stoichiometry. For low-confidence regions (LSCO, “Other”), complex doping patterns and non-standard materials require additional training data or architectural improvements.

Could 868,882 candidates find holdout matches by chance? The search generated $\sim 19,300$ candidates per target across $\sim 121,650$ Z-vectors (many candidates are duplicates or near-duplicates). A naive estimate of the chemical composition space is instructive: with 118 elements and typical superconductor formulas containing 3–7 elements with fractional stoichiometries, the combinatorial space of chemically plausible compositions is astronomically large ($> 10^{10}$ even restricting to known

superconductor-like patterns). Finding 17 exact matches and 32/45 above 0.99 similarity from $\sim 20K$ unique candidates per target by random sampling would require extraordinary luck. More concretely, the search is not random: it is seeded from element-anchored Z-vectors of chemically similar training materials and explores the *local* latent neighborhood around those seeds. The high recovery rates therefore reflect the latent space’s organization—chemically similar materials map to nearby Z-vectors, and the decoder produces chemically coherent outputs in those regions. The fact that zero extra elements were hallucinated (the model never adds elements that don’t belong) further argues against chance: a random walk through Z-space would produce arbitrary element combinations, not consistently correct element sets with only minor stoichiometric errors.

Evaluating a generator, not a regressor. The 45-target holdout evaluation deserves explicit methodological justification. In standard T_c prediction work, models are evaluated by a train/validation/test split (typically 80/10/10), with test-set MAE or R^2 as the metric. A 20% test split from 23,451 SC samples would yield $\sim 4,700$ test materials—statistically powerful but testing only whether the model can *predict a number* from a composition. The present evaluation asks a fundamentally harder question: can the model *generate* a never-seen composition from latent space exploration alone, with no target information as input? This is not a regression test but a combinatorial search problem. Each of the 45 targets is a specific needle in a haystack of $> 10^{10}$ chemically plausible compositions, and the model must find it by exploring $\sim 20,000$ candidates in a 2048-dimensional space. Finding 17/45 exact matches under these conditions is substantially more demanding than predicting T_c for 4,700 known compositions, because the model must simultaneously get the element set, all stoichiometric fractions, the T_c , and the SC classification correct—all without seeing any target data.

The difficulty is compounded by a fundamental aspect of superconductivity: the phenomenon arises from multiple mutually exclusive physical mechanisms. BCS/Allen-Dynes phonon-mediated pairing governs elemental and A15 superconductors (35; 32); MgB_2 requires a two-gap strong-coupling framework (40); cuprate superconductivity involves *d*-wave pairing mediated by spin fluctuations (36); iron pnictides exhibit $s_{\pm}$ -wave pairing from inter-band nesting (41); and heavy-fermion and organic superconductors involve still other mechanisms. The model must implicitly learn *which mechanism applies* from composition alone—without explicit labels for the pairing symmetry, mediating boson, or order parameter—and correctly predict T_c under the appropriate physical framework. The holdout set spans all of these mechanisms: conventional BCS metals, MgB_2 variants, four cuprate sub-families, iron pnictides, and miscellaneous compounds. That the model achieves 95.6% recovery across this mechanistic diversity, with family-specific T_c accuracy (Table 10), provides evidence that the latent space has organized itself into a multi-theory representation.

The small holdout size is a deliberate design choice: every superconductor reserved for testing is one the generator cannot learn from, and a generator that must span these diverse mechanisms benefits from maximal training coverage. The 45 targets (5 per family $\times$ 9 families) provide sufficient diversity to evaluate generalization across the field while preserving training data density in under-represented families. Future external validation can extend this evaluation to superconductors in databases not used during training (e.g., SuperCon2, SODNet), providing arbitrarily large test sets without sacrificing training coverage.

Limitations. The $R^2 = 0.9999$ is on training data; the generative evaluation uses 45 holdout targets, which demonstrates latent-space generalization across families and mechanisms but with limited per-family sample size. The composition-only input means structural variables can only be inferred statistically. The dominant error mode is stoichiometric fraction imprecision (98% of all errors), and equal site-sharing patterns (e.g., $La_{0.5}Gd_{0.5}$) can collapse to single-element occupancy. Latent torsion between the decoder and T_c head manifolds causes two targets to exhibit > 14 K T_c error despite high compositional similarity. No physical constraints (charge neutrality, thermodynamic stability) are enforced during generation.

Future data integration. Several additional databases have been staged but not yet integrated: SuperCon2 ($\sim 40K$ NLP-extracted records from 37,700 papers), 3DSC (1,578 superconductors with crystal structures (7)), SODNet (curated with verified T_c), and JARVIS hydrides (DFT-predicted). Rather than training on all new data uniformly, the model’s current competence suggests a curriculum strategy: generate candidates from the new databases, identify which ones the model struggles to reproduce or predict accurately, and selectively train on those failure cases. This targeted augmentation—already demonstrated in the iterative addition of rare-element non-SC materials during initial training—would maximize the marginal value of each new training sample. Incorporating 3DSC structural data

could address the hidden variable problem, but must be done carefully: structural data availability is inhomogeneous across families, and selectively including it risks introducing training bias.

Future modeling directions. The most important next step is *mechanistic interpretability*: extracting the effective theory that MAGE has learned as explicit mathematical equations. The model’s weights, latent space geometry, and input-output mappings implicitly encode a multi-family theory of superconductivity that produces accurate T_c predictions and chemically valid formulas across 14 material families. The goal of the next research phase is to fit these implicit relationships—the T_c head’s learned function of $\mathbf{z}$, the family-specific gradient profiles, the decoder’s conditional generation rules—into closed-form expressions that a physicist can interpret, test, and extend. This would transform the model from a predictive tool into a source of new theoretical insight: equations derived from the model’s learned representation could reveal which compositional features drive T_c in each family, what the latent torsion encodes about structural dependence, and whether the smooth transitions between families in latent space correspond to known or novel physical relationships. Additional directions include reducing latent torsion through explicit manifold alignment losses (e.g., penalizing T_c error at Z-vectors that produce high-similarity formulas) or self-consistency training; incorporating physical constraints (charge neutrality, stability) in generation; external validation on superconductors from databases not used in training (SuperCon2, SODNet) to provide large-scale generalization evidence without sacrificing training data; and addressing the stoichiometric precision bottleneck through constrained decoding or post-hoc normalization. The observation that later, larger models did not outperform the epoch 3292 checkpoint merits systematic investigation of the capacity–quality tradeoff in autoregressive materials generation.

7 Conclusion

I developed a multi-task sequence encoder-decoder that achieves $R^2 = 0.9999$ on T_c prediction—a $27\times$ error reduction over a composition-only baseline—and generalizes to 45 never-seen superconductors. A Z-space search generating 868,882 candidate formulas via element-anchored latent space exploration recovers exact compositional matches for 17/45 holdout targets (37.8%) and the correct element set for 43/45 (95.6%), with zero hallucinated elements and 98% of errors limited to stoichiometric fraction imprecision. This represents a $5.7\times$ improvement over roundtrip encode-decode (3/45 exact, mean similarity 0.944), demonstrating that the learned latent space contains never-seen superconductors at navigable locations. All 45 targets were correctly classified as superconductors (100%), and T_c predictions achieved a median error of 0.40 K without receiving any target information. The 2048-dimensional latent space self-organizes into a physically meaningful manifold with family-specific structure. Unlike prior two-stage generative approaches, the unified architecture supports both formula generation and multi-property prediction through a single shared latent space. The model has learned what I characterize as a 2048-parameter effective theory of superconductivity—a single representation spanning 14 material families that encodes family boundaries, the SC/non-SC phase boundary, and continuous T_c gradients within a geometrically organized manifold.

Code is available at (31).

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